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A

Power – 12V, solar and USB inputs, 18650 charger, rails

File: power.kicad_sch

ESP32, buttons, status LEDs, buzzer, I2C

File: mcu.kicad_sch

LoRa, Ethernet, cellular (all optional)

File: comms.kicad_sch

B

AgriNode HUB–1 – universal gateway hub (single–sided PCB)

Variants by assembly option: HUB–W (Wi–Fi), HUB–L (+LORA), HUB–C (+CELL), GW–LAN (+LORA +ETH); BUZ optional

ESP32–WROOM–32E | LoRa Ra–02 433 MHz | SIM800C 2G | ENC28J60 10BASE–T | I2C expansion header

Power: 12 V jack/terminal, solar 6–24 V, USB–B 5 V, 18650 cell with TP4056 charger + NTC, power path, MIC29302 3.3 V

6x PL9823 addressable RGB status LEDs, RESET and BOOT/PAIR buttons, buzzer

Source of truth: design/hub.py – regenerate with build.py

C

D

E

Generated by tools/kigen from the Python design source – edit the source, not this file

AgriNode

Sheet: /

File: agrinode_hub.kicad_sch

Title: AgriNode HUB–1 Gateway – Overview

Size: A3

Date: 2026–09–27

Rev: A

KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4

Id: 1/4

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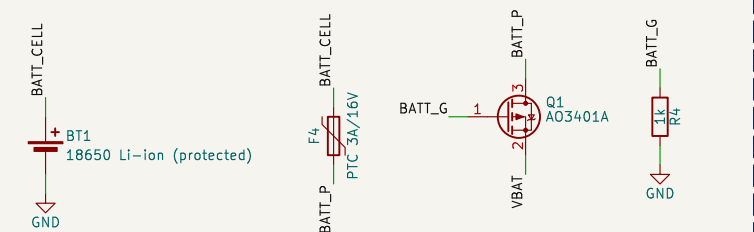
6

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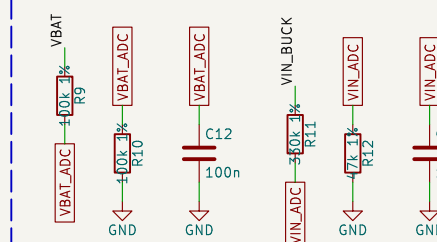
each input fused and reverse-polarity protected; any combination may be connected

18650 cell: holder, fuse, reverse-insertion protection

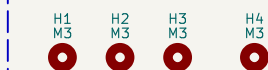


Supply monitoring (ADC1)

VBAT: 100k/100k VIN: 330k/47k (24 V $\rightarrow$ 3.0 V)



Mounting holes (M3)

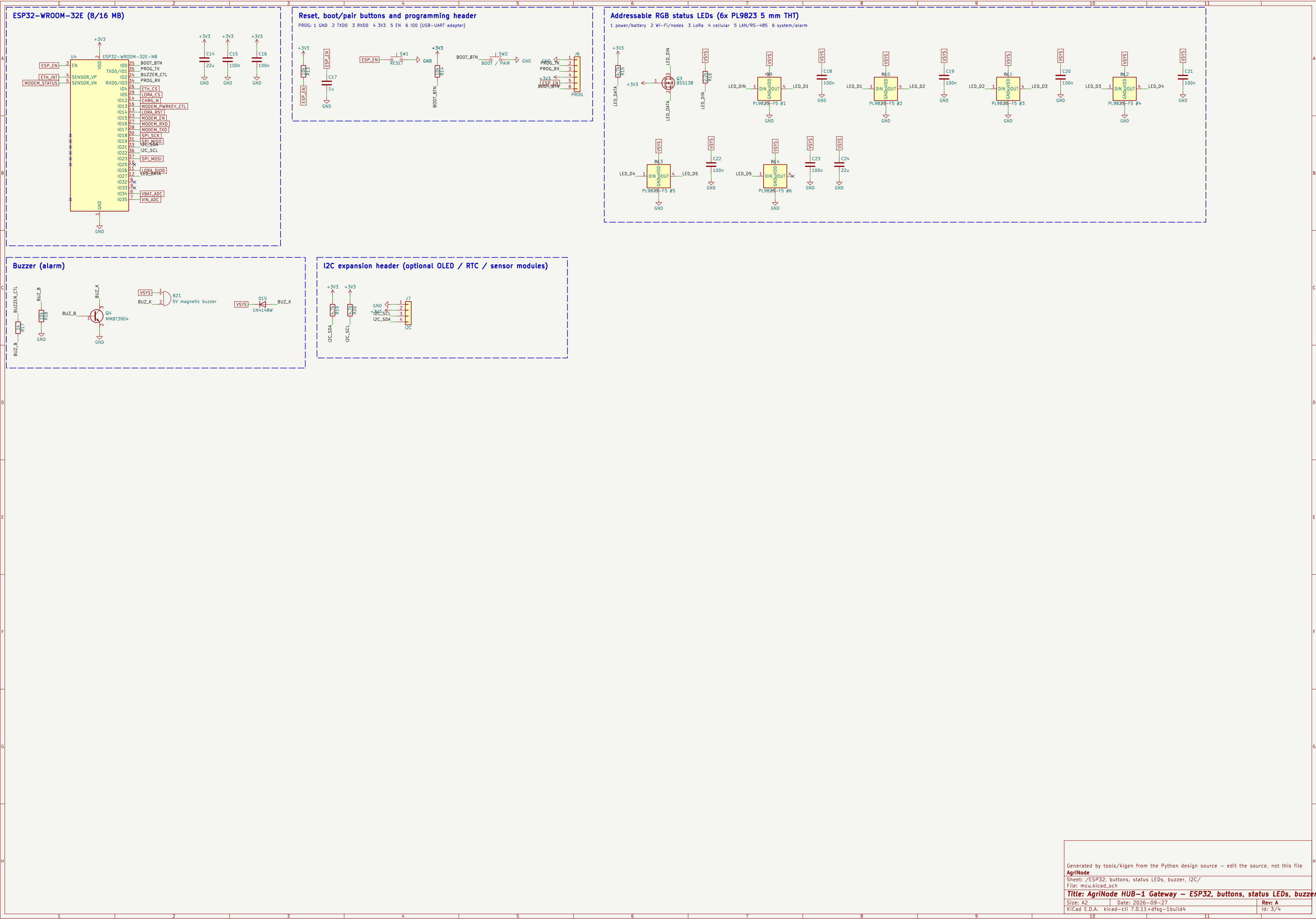


AgriNode

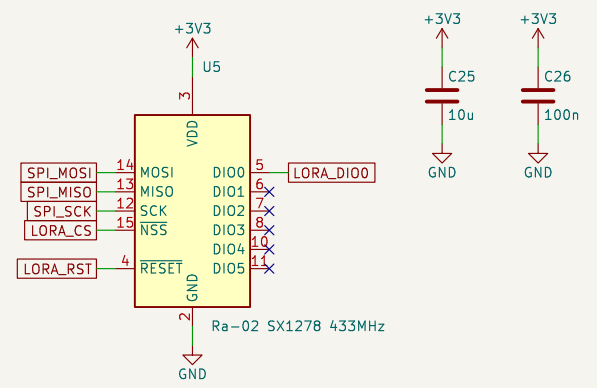
Title: AgriNode HUB-1 Gateway – Power – 12V, solar and USB inputs, 1865

Size: AS	Date: 2020-09-27	REV: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/4

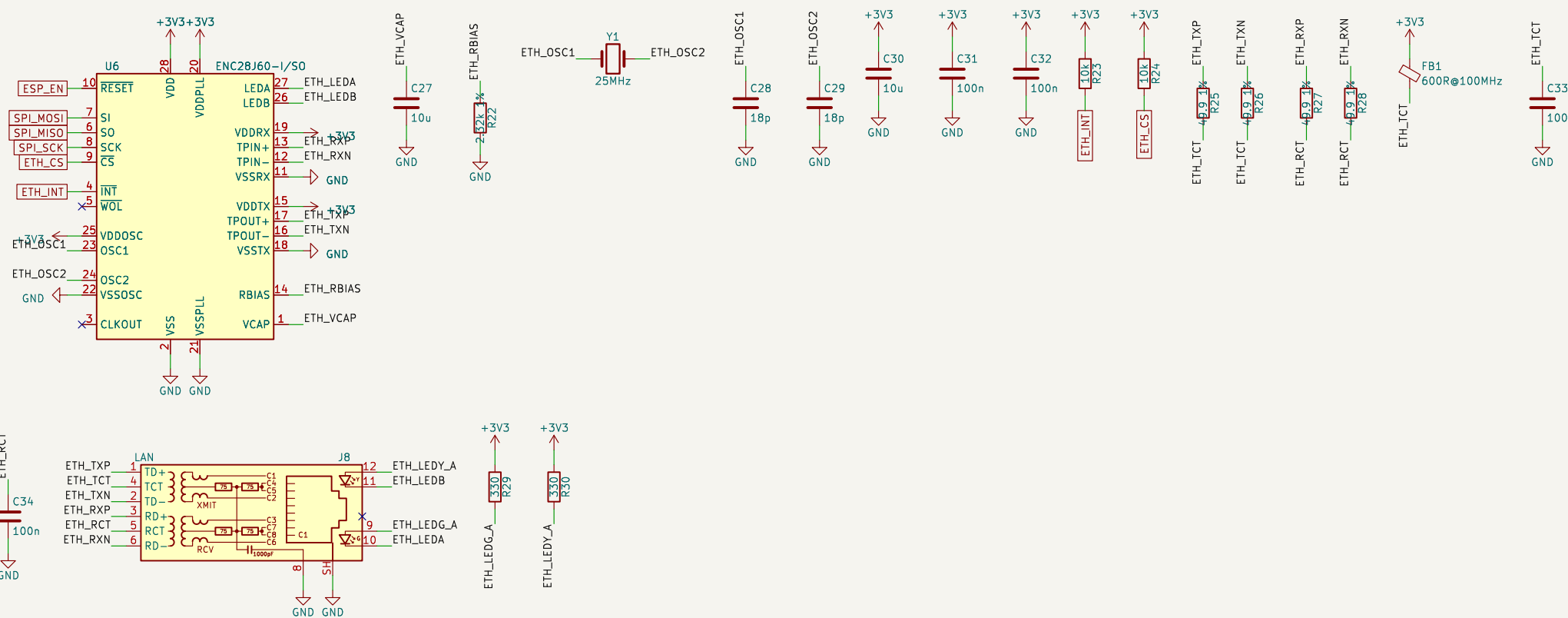
	7	8	F
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LoRa 433 MHz (Ai-Thinker Ra-02, IPEX antenna)



Ethernet 10BASE-T (ENC28J60 + HR911105A MagJack)



Cellular 2G (SIM800C): GPRS, SMS, voice alarm calls

